



Khwaja Yunus Ali University

Name of the Department: Computer Science and Engineering

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Course Title: Electrical Circuit

Assingment No.: 02

Name of the Assignment : Application oriented relation analysis of AC and DC circuits.

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Instructor Signature & Date

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Definition of DC (Direct Current)

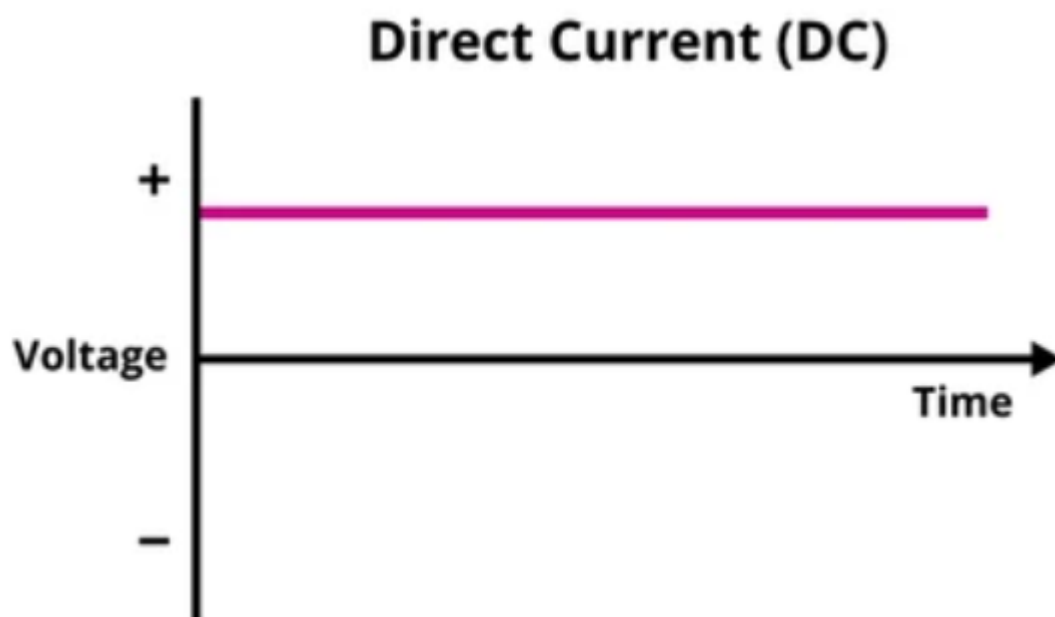
Direct Current (DC) is the unidirectional flow of electric charge. In a DC circuit, electrons move in one direction—**from negative to positive terminal**. The voltage and current are constant over time.

Examples:

- Batteries
- Solar cells
- Power banks

DC Signal Graph:

The graph of DC voltage or current is a **straight horizontal line**, indicating a **constant value**.



Definition of AC (Alternating Current)

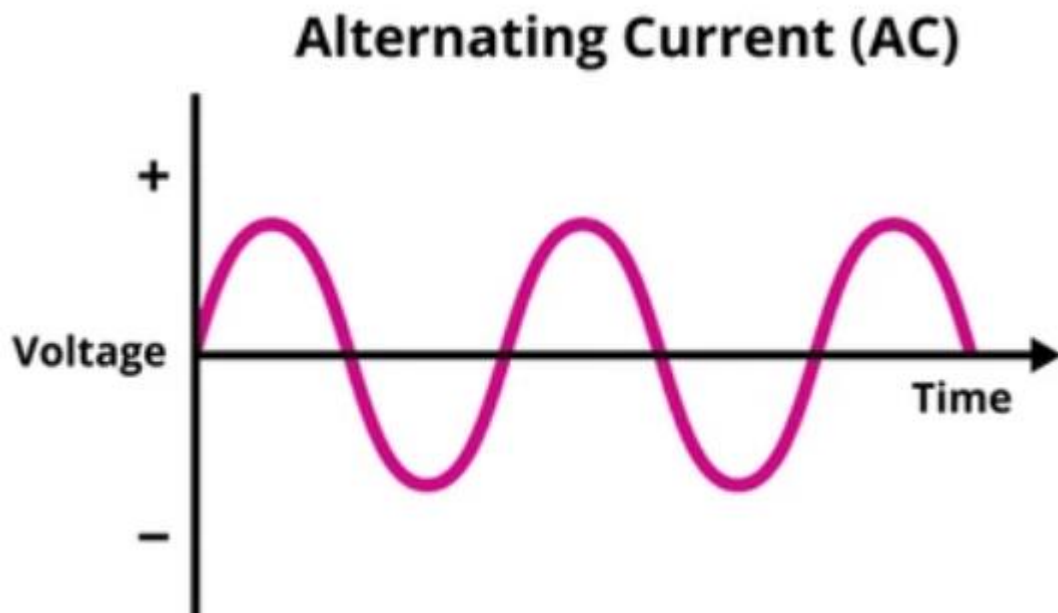
Alternating Current (AC) is an electric current that **reverses direction periodically**. The voltage also alternates between positive and negative values in a sinusoidal pattern.

Examples:

- Household power supply (like 220V/50Hz)
- Generators
- AC power adapters

AC Signal Graph:

The graph of AC voltage or current is a **sine wave**, which shows how the voltage **alternates** over time.



Similarities Between AC and DC

◇ 1. Both Involve the Flow of Electrons

- AC and DC both represent the flow of electric charge (electrons) through a conductor.

2. Both Can Power Electrical Devices

- Devices like lights, fans, and motors can run on either AC or DC (with appropriate converters or design).

3. Both Obey Ohm's Law

- The relationship $V=IR$ (voltage = current $\times$ resistance) applies to both AC and DC circuits.

4. Both Can Be Converted into Each Other

- **Rectifiers** convert AC to DC.
- **Inverters** convert DC to AC.

5. Both Experience Power Loss

- Both AC and DC systems suffer from power loss due to resistance in the wires (I^2R losses).

6. Both Can Be Measured Using Electrical Instruments

- Instruments like voltmeters, ammeters, and multimeters can measure both AC and DC values.

Differences Between AC and DC

Feature	AC (Alternating Current)	DC (Direct Current)
Direction of Flow	Changes direction periodically	Flows in one direction only
Waveform	Sine wave (can also be square or triangular)	Straight horizontal line
Source	Power plants, generators	Batteries, solar cells

Feature	AC (Alternating Current)	DC (Direct Current)
Frequency (Hz)	50 Hz or 60 Hz (varies by country)	0 Hz (no frequency)
Transmission	Efficient for long distances	Not efficient for long distances
Voltage Level	Alternates between positive and negative	Constant and steady
Conversion	Can be converted to DC using a rectifier	Can be converted to AC using an inverter
Usage	Homes, industries, appliances	Electronics, mobile devices, vehicles
Current Reversal	Yes	No
Power Loss	Lower due to easy transformation (high voltage)	Higher over long distances

Applications of DC Circuits

DC circuits are used in situations where a constant voltage or current is required.

Common Applications:

1. **Electronic Devices:** Laptops, mobile phones, tablets, and calculators.
2. **Battery-Powered Equipment:** Flashlights, remote controls, electric toys.
3. **Electric Vehicles (EVs):** Most EV systems run on DC supplied by batteries.
4. **DC Motors:** Used in small fans, drones, electric scooters, and robotics.
5. **Solar Power Systems:** Solar panels generate DC electricity.
6. **Portable Power Banks:** Store and supply DC to charge devices.
7. **Embedded Systems:** Microcontrollers and sensors operate on DC supply.

Applications of AC Circuits

AC circuits are used where electricity needs to be transmitted or distributed over long distances.

Common Applications:

1. **Household Power Supply:** AC is used in home wiring to power lights, fans, TVs, etc.
2. **Industrial Machinery:** Large AC motors used in factories and workshops.

3. **HVAC Systems:** Heating, ventilation, and air conditioning systems.
4. **Public Utility Grids:** AC is used for high-voltage power transmission.
5. **Appliances:** Refrigerators, washing machines, air conditioners.
6. **Transformers:** AC is essential for voltage transformation using transformers.

Application of AC and DC Circuits for Computer Engineers

Computer engineers use both AC and DC circuits in various hardware and system designs.

DC Applications:

- **Powering Internal Components:** CPUs, RAM, SSDs, and GPUs all operate on low-voltage DC.
- **Motherboards and ICs:** Work exclusively with DC power.
- **Battery-Operated Devices:** Laptops, tablets, and embedded systems use DC circuits.
- **Microcontrollers & Sensors:** Used in IoT and automation projects.

AC Applications:

- **Power Supply Units (PSUs):** Convert AC from wall sockets into DC for computer components.
- **Server Rooms and Data Centers:** AC circuits are used to distribute main power.
- **UPS Systems (Uninterruptible Power Supply):** Use both AC (input/output) and DC (battery backup).
- **Cooling and HVAC Systems:** Maintain environment of computing hardware.

Personal Experience with AC and DC Quantities

Phenomenon Description:

One day, during a power outage at home, I decided to charge my phone using a **power bank** while also setting up a **desktop computer** on an **Uninterruptible Power Supply (UPS)** to continue my work. This incident made me clearly observe and understand the practical difference and application of **DC and AC**.

DC Usage:

The **power bank** I used was charged beforehand and supplied **DC power** to my mobile phone. The phone's charging circuit works on **low-voltage DC (5V)**, which was directly supplied from the power bank. This helped me stay connected even without main electricity.

AC Usage:

At the same time, the **UPS** supplied **AC power** to my desktop computer. Even though the internal components of the computer work on DC, the UPS initially delivers **AC voltage (220V)**, which the **power supply unit (SMPS)** of the computer later converts into different DC voltages like 3.3V, 5V, and 12V.

What I Learned:

This scenario helped me understand:

- **DC is essential for low-voltage portable electronics.**
- **AC is required for high-voltage power distribution and powering large devices.**
- How **conversion between AC and DC** is critical in everyday electronic systems.